Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *GLUE* | ***Context:*** *Luxembourgish Public Administration Civil Servants — GLUE Assessment*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GLUE inputs are standardized written English text in a single script with consistent orthography, formatted as single sentences or sentence pairs [Q4, Q15]. The deployment input is text in three languages with frequent code-switching at lexical, phrasal, and sentence level [WEB-11], non-standardized Luxembourgish spelling where 'large amount of orthographic variation' and homograph density are documented challenges [WEB-12, WEB-15], and a register requiring trilingual tokenization and normalization that English NLP pipelines do not provide [WEB-2, WEB-13, WEB-14]. GLUE's preprocessing transformations (recasting SQuAD into NLI [Q42], converting Winograd [Q52]) are English-specific and provide no transferable methodology for non-standardized multilingual signals. IF is HIGH priority and the gap is fundamental.

ID	Evidence
Q4	"GLUE does not place any constraints on model architecture beyond the ability to process single-sentence and sentence-pair inputs and to make corresponding predictions." (p.1)
Q15	"In summary, we offer: (i) A suite of nine sentence or sentence-pair NLU tasks, built on established annotated datasets and selected to cover a diverse range of text genres, dataset sizes, and degrees of difficulty. (ii) An online evaluation platform and leaderboard, based primarily on privately-held test data. The platform is model-agnostic, and can evaluate any method capable of producing results on all nine tasks. (iii) An expert-constructed diagnostic evaluation dataset. (iv) Baseline results for several major existing approaches to sentence representation learning." (p.2)
Q42	"We convert the task into sentence pair classification by forming a pair between each question and each sentence in the corresponding context, and filtering out pairs with low lexical overlap between the question and the context sentence." (p.4)
Q52	"To convert the problem into sentence pair classification, we construct sentence pairs by replacing the ambiguous pronoun with each possible referent." (p.4)
WEB-11	LuxBorrow corpus with token-level LID and code-switching annotation across LU/DE/FR/EN — confirms code-switching as systematic input characteristic (https://arxiv.org/html/2603.10789)
WEB-12	Neural Text Normalization for Luxembourgish — best ByT5 base achieves ~78.8% accuracy; documents 'large amount of orthographic variation' (https://arxiv.org/html/2412.09383v1)
WEB-15	spellux Python package documents homograph density and orthographic variation as ongoing Luxembourgish NLP challenges (https://github.com/questoph/spellux)
WEB-2	CTIE operates MyGuichet.lu under Ministry for Digitalisation (https://european-language-equality.eu/wp-content/uploads/2022/03/ELE___Deliverable_D1_24__Language_Report_Luxembourgish_.pdf)
WEB-13	LuxemBERT (2022) outperformed mBERT — confirms standardized English models are inadequate baselines for Luxembourgish (https://aclanthology.org/2022.lrec-1.543.pdf)
WEB-14	https://sirajzade.github.io/papers/CRC_industrial_paper_84.pdf

Strengths

- GLUE imposes minimal architectural constraints, requiring only sentence/sentence-pair input [Q4]; this format-agnostic design means a custom Luxembourgish benchmark could in principle adopt the same I/O contract while substituting language-appropriate content.

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Q4	"GLUE does not place any constraints on model architecture beyond the ability to process single-sentence and sentence-pair inputs and to make corresponding predictions." (p.1)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions are fundamentally mismatched. GLUE inputs are standardized English text [Q4, Q18]; deployment inputs are trilingual lb/fr/de code-switched text with non-standardized Luxembourgish orthography. The LuxBorrow corpus [WEB-11] documents systematic token-level language switching, and the best Luxembourgish neural normalizer achieves only ~78.8% accuracy [WEB-12], indicating ~21% residual orthographic noise that GLUE-trained models would never have encountered.

IF-2: Luxembourg's digital infrastructure for capturing citizen correspondence (MyGuichet.lu via CTIE [WEB-2]) supports text input and is technically capable of feeding any text-based pipeline; the infrastructural gap is not capture but processing — Luxembourgish NLP tooling (LuxemBERT [WEB-13], LuNa [WEB-14], spellux [WEB-15], ltz-E1 [WEB-10]) is emerging but limited compared to the standardized English NLP stack GLUE assumes.

IF-3: Domain-specific form differences include: (a) trilingual tokenization requirements that English tokenizers do not satisfy; (b) orthographic normalization needs specific to Luxembourgish [WEB-12, WEB-15]; (c) handling of contact-linguistic phenomena [WEB-11]. GLUE's transformations [Q42, Q49, Q52] address none of these.

IF-4: External-validity violation is severe: a model evaluated only on standardized English text input has zero exposure to the non-standard, code-switched, multilingual input form characteristic of Luxembourgish citizen correspondence. The 409-example dataset sample contains no non-English tokens, confirming the form mismatch empirically (DATASET-Concern 1).

ID	Evidence
Q4	“GLUE does not place any constraints on model architecture beyond the ability to process single-sentence and sentence-pair inputs and to make corresponding predictions.” (p.1)
Q18	“GLUE is centered on nine English sentence understanding tasks, which cover a broad range of domains, data quantities, and difficulties.” (p.3)
WEB-11	LuxBorrow corpus with token-level LID and code-switching annotation across LU/DE/FR/EN — confirms code-switching as systematic input characteristic (https://arxiv.org/html/2603.10789)
WEB-12	Neural Text Normalization for Luxembourgish — best ByT5 base achieves ~78.8% accuracy; documents 'large amount of orthographic variation' (https://arxiv.org/html/2412.09383v1)
WEB-2	CTIE operates MyGuichet.lu under Ministry for Digitalisation (https://european-language-equality.eu/wp-content/uploads/2022/03/ELE___Deliverable_D1_24__Language_Report_Luxembourgish_.pdf)
WEB-10	ltzGLUE provides Luxembourgish sentiment (positive/neutral/negative) but not in administrative register (https://arxiv.org/abs/2604.17976)
WEB-13	LuxemBERT (2022) outperformed mBERT — confirms standardized English models are inadequate baselines for Luxembourgish (https://aclanthology.org/2022.lrec-1.543.pdf)
WEB-14	https://sirajzade.github.io/papers/CRC_industrial_paper_84.pdf
WEB-15	spellux Python package documents homograph density and orthographic variation as ongoing Luxembourgish NLP challenges (https://github.com/questoph/spellux)
Q42	“We convert the task into sentence pair classification by forming a pair between each question and each sentence in the corresponding context, and filtering out pairs with low lexical overlap between the question and the context sentence.” (p.4)
Q49	“We convert all datasets to a two-class split, where for three-class datasets we collapse neutral and contradiction into not entailment, for consistency.” (p.4)

Q52	"To convert the problem into sentence pair classification, we construct sentence pairs by replacing the ambiguous pronoun with each possible referent." (p.4)

Information Gaps

- Exact distribution of language proportions and code-switching density in actual MyGuichet.lu citizen correspondence is not publicly documented and would require CTIE access.

Requires Expert Verification

- Whether a normalization preprocessing step (e.g., spellux or ByT5 normalizer) is sufficient to recover GLUE-trained model performance on Luxembourgish input, or whether retraining is required, needs empirical validation.

Remediation

Gap 1: GLUE assumes standardized English text; deployment requires trilingual lb/fr/de tokenization and Luxembourgish orthographic normalization.

Recommendation: Integrate a preprocessing pipeline using spellux [WEB-15] or ByT5-based normalization [WEB-12] for Luxembourgish input, combined with multilingual tokenization (e.g., LuxemBERT [WEB-13] or LuNa [WEB-14] tooling). Empirically validate residual error rates and propagation to downstream tasks before deployment.

ID	Evidence
WEB-15	spellux Python package documents homograph density and orthographic variation as ongoing Luxembourgish NLP challenges (https://github.com/questoph/spellux)
WEB-12	Neural Text Normalization for Luxembourgish — best ByT5 base achieves ~78.8% accuracy; documents 'large amount of orthographic variation' (https://arxiv.org/html/2412.09383v1)
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